INCH-POUND

MIL-A-53129
16 September 1992

MILITARY SPECIFICATION

ALARM-MONITOR GROUP OA-9431/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes an Alarm-Monitor Group, which when connected and operated with an IBM PC/AT compatible (ISA Bus) personal computer becomes the primary notification device for the Joint-Services Interior Intrusion Detection System (J-SIIDS). The Alarm-Monitor Group consists of both commercial and military designed components. Military specifications called out in this document only apply to the military designed hardware (Converter-Multiplexer Assembly). Compliance with the functional requirements can only be demonstrated when the commercial hardware and military hardware are assembled and tested in accordance with this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA, 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-81997	- Pouches, Cushioned, Flexible, Electrostatic-Free Recloseable, Transparent.
MIL-T-52877	- Transmission Group Data: Transmitter, Data: T-1257/FSS-9(V) and Receiver, Data: R-1861/FSS-9(V)

STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-2000	- Soldering.
MIL-STD-45662	- Calibration System Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.

(Unless otherwise specified, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

ME

TA13229E8200	- Alarm-Monitor Group OA-9431/FSS-9(V).
13229E8140	- Converter Multiplexer Assembly
13229E8149	- Communications Circuit Card Assembly.
13229E8340	- Uninterruptible Power Supply.

(Copies of drawings required by the contractor in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-JIC, Fort Belvoir, VA 22060-5606.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	- Standard Practice for Commercial Packaging.
ASTM D 3953	- Strapping, Flat Steel and Seals.
ASTM D 4675	- Selection and Use of flat Strapping Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-485	- Recommended Standard for Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC, 20006.)

AMERICAN WELDING SOCIETY (AWS)

AWS D 9.1	- Sheet Metal Welding Code.

(Application for copies should be addressed to American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take

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precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Alarm-Monitor Group (hereinafter called AMG) shall be in accordance with top assembly TA13229E8200, and as specified herein. The AMG, when used in conjunction with commercial computer components, serves as the annunciator for the J-SIIDS. It consists of both hardware and software items.

3.1.1 Major components. The AMG shall contain the following major components:

a. Alarm-Monitor Program Diskettes (diskettes #1 and #2), title "AMG PC software, 13229E8349"
b. Converter-Multiplexer Assembly (CMA), 13229E8140
c. Communications circuit card assembly (CCCA), drawing 13229E8149
d. Uninterruptible Power Supply (UPS) with batteries, drawing 13229E8340

When specified (see 6.2), any of the major components listed above shall be furnished separately.

3.1.2 Function. The AMG, when used in conjunction with an MS-DOS, ISA bus personal computer, a color monitor, and a printer, shall monitor and display remote area status information from up to 64 J-SIIDS Control Units, C-9412/FSS-9(V). Information displayed from each control unit shall be as follows: Alarm/No Alarm; Access/Secure; AC Power/AC Power Fail (Battery Power). Information is transmitted over telephone lines for up to ten miles to the AMG by the J-SIIDS Data Transmitters, T-1275/FSS-9(V), conforming to MIL-T-52877. This information shall be interpreted and displayed in accordance with the requirements herein. In addition the AMG, when used in conjunction with an MS-DOS, ISA bus personal computer, a color monitor, and a printer, shall monitor and display the status of the AMG's tamper switches and the status of the UPS power source.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the AMG. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provision of this specification.

3.3.1 Material deterioration prevention and control. The AMG shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that

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may be encountered in any of the applicable storage and operating environments to which the AMG may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the AMG or major components when furnished separately may be newly fabricated from recovered materials to the maximum extent practicable, provided the AMG meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the AMG.

3.4 Environmental conditions. The AMG shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.5.1), performance degradation, or malfunction.

3.4.1 Operational.

a. Temperature range of 68 to 84 °F.
b. Humidity up to 80 percent relative.

3.5 Detailed technical requirements. The AMG shall perform as specified herein under any of the operating test conditions specified herein when the hardware components are properly connected with the specified computer equipment and the programs on the AMG diskettes are properly installed.

3.5.1 CMA. The CMA shall receive coded status data from up to 64 data transmitters. This data shall be transmitted to the CCCA and shall be correctly interpreted by the alarm-monitor program to display the current status at the data transmitter inputs. Changes at the data transmitter inputs shall be reported by the alarm-monitor program within 5 seconds.

3.5.1.1 Status data reporting. The alarm-monitor program shall receive data from the data transmitter through the CMA and CCCA, and it shall display the data as described in section 3.5.1.1.1 through 3.5.1.1.4. The data transmitted shall correspond to the states at the four data transmitter inputs. The four inputs are an alarm status input, mode switch status input, ac power status input, and resynchronization input. Any change at a data transmitter input from the normal state to the non-normal shall be displayed as specified herein and blinking, and a beeper shall sound. Changes indicated by the alarm-monitor program shall be acknowledgeable by pressing <F1> on the keyboard. Acknowledging the change shall silence the beeper and stop the indicator from blinking.

3.5.1.1.1 Alarm status. The normal state at the data transmitter alarm status input is closed with respect to signal ground and is defined as no alarm. The

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non-normal state is defined as alarm. The alarm-monitor program shall display nothing for the normal state. In the non-normal state, the alarm-monitor program shall display the text "ALM" in white-on-red. Changes from the non-normal state to the normal state shall be displayed as "ALM" in gray-on-red until reset by pressing <F2>. Resetting the "ALM" indication when the alarm status input is in the normal state shall cause the indication to be removed from the display.

3.5.1.1.2 Mode switch status. The normal state at the data transmitter mode switch status input is open with respect to signal ground and is defined as secure. The non-normal state is defined as access. The alarm-monitor program shall display the text "SEC" in green for the normal state. In the non-normal state the alarm-monitor program shall display the text "ACC" in yellow. A change from access to secure shall also be indicated as blinking and cause the beeper to sound.

3.5.1.1.3 AC power status. The normal state at the data transmitter ac power status input is closed with respect to signal ground and is defined as ac power present. The non-normal state is defined as ac power fail. The alarm-monitor program shall display the text "AC" in green for the normal state. In the non-normal state the alarm-monitor program shall display the text "BAT" in yellow. A change from ac power fail to ac power present shall also be indicated as blinking and cause the beeper to sound.

3.5.1.1.4 Resynchronization. The normal state at the data transmitter resynchronization input is open with respect to signal ground. There is nothing displayed by the alarm monitor program when the data transmitter is in the normal state. The non-normal state occurs when the input is grounded and opened again, initiating the data transmitter to send a unique message which signals the receiver to resynchronize. The printer shall print the message "Resynch-M".

3.5.1.2 Communication failure reporting. The alarm-monitor program shall report any failures of the communication between the data transmitter and the CMA. It shall also report any failures of the communication between the CMA and the CCCA. Communication failures shall be reported by the alarm-monitor program within 5 seconds.

3.5.1.2.1 Communication failure between data transmitter and CMA. Failure of the data transmitter and CMA to communicate properly shall result in the alarm-monitor program indicating "COM" in white-on-red conforming to the display of an alarm (see 3.5.1.1.1). Corresponding status indications for that data transmitter shall be changed to display in gray. The printer shall print the message "COM-FAIL-*xyz*" where *x*, *y*, and *z* shall each be a number "1" or "0". The number "1" in the *x* position shall indicate a loss of synchronization; a "1" in the *y* position shall indicate a loss of readable data; and a "1" in the *z* position shall indicate a low signal level (see 3.5.2). A "0" is the normal condition indication. Changes from the non-normal state to the normal state shall be displayed as "COM" in gray-on-red until reset by pressing <F2>. The other status indicators shall be restored to the color representing the current state (see 3.5.1.1). The printer shall print "COM-norm-000". Resetting the "COM" indication when communication has returned to the normal state shall cause the indication to be removed from the display.

3.5.1.2.2 Communication failure between CMA and CCCA. Loss or partial loss of the communication between the CMA and the CCCA shall result in the alarm-monitor

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program displaying "COM" in white-on-red for each of eight modules which has lost communication. The printer shall print the message "COM-FAIL" for each of the failed modules. Changes from the non-normal state to the normal state shall be displayed as "COM" in gray-on-red until reset by pressing <F2>. Resetting when communication has returned to the normal state shall cause the "COM" indication to be removed from the display.

3.5.1.3 Automatic synchronization. The alarm-monitor program shall automatically synchronize with the data streams of up to 64 data transmitters converted by the CMA. Synchronization shall occur within 5 minutes of the restoration of the communication. At initial start up of the program, the text "SYNC" shall be indicated in light blue to indicate the out of synchronization state. Any loss of synchronization after initial start up shall be indicated as a "COM" (see 3.5.1.2.1). Automatic synchronization shall be indicated by the change of the other status indicators of the data transmitter to the color representing the current state at the inputs (see 3.5.1.1). At start up, the "SYNC" indicator shall disappear. When "COM" was indicated, the "COM" indication displays the return to normal state (see 3.5.1.2.1). The printer shall print the message "Resynch-A".

3.5.1.4 CMA input sensitivity. The AMG shall perform as specified herein when the input signal level at the phone line input is -33 ± 2 dBm or greater. When the signal level falls to this level or below, a communication failure shall be displayed (see 3.5.1.2.1).

3.5.1.5 Tamper switch operation. Tamper switches within the CMA shall have normally closed contacts. The switches shall be mounted and adjusted such that their contacts shall open on not less than 1/8 inch nor more than 1/4 inch of door, drawer, panel, or ventilation panel movement. Tamper alarms from the CMA shall be reported by the alarm-monitor program within 3 seconds of any one or more tamper switch opening. CMA tamper alarms are indicated by the text "TAMPER" in white-on-red and conforming to alarm conventions (see 3.5.1.1.1)

3.5.1.6 UPS status. The UPS status shall be transmitted by the CMA to the alarm-monitor program. The alarm-monitor program shall display changes in the UPS status within 3 seconds of the changes. The normal state of the UPS status output is closed with respect to signal ground and is defined as operating on facility ac power. The non-normal state is defined as operating on backup battery power. The normal state is indicated by the text "AC POWER" in green and the non-normal state is indicated by "BATTERY" in yellow.

3.5.1.7 Presence of data receiver CCAs. The CMA shall detect and transmit the presence or absence of all 64 data receiver CCAs to the CCCA. The alarm-monitor program shall display this information. The normal state is defined as present and is displayed by the alarm-monitor program as "Y". The non-normal state is defined as not-present and is displayed as ".".

3.5.1.8 Tamper and UPS status transmitted by all modules. The status of the tamper switches and the UPS shall be transmitted to the CCCA by each of the eight CMA modules.

3.5.1.9 Reporting capacity. The alarm-monitor program shall report status changes initiated simultaneously on all 64 CMA inputs within 12 seconds.

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3.5.1.10 Automatic startup. The CMA shall start operating automatically on power-up. The alarm-monitor program shall automatically start operating when the computer on which it is installed is powered on.

3.5.2 CCCA. The CCCA shall be physically installed within the computer. Operation of the AMG and computer shall be as specified herein when the CCCA is set to the following configuration parameters: COM4 (address 2E8 Hex); IRQ4 (interrupt 4); RTS and CTS loopback; receive clock is from UART baud rate generator; full duplex operation; direct clock; and channel A enabled.

3.5.3 UPS. In the event of removal of input AC power, the UPS shall provide an uninterrupted supply of power and shall operate the AMG and computer equipment for a minimum of 4 hours after loss of input AC power.

3.5.4 Alarm-monitor program diskettes. The alarm-monitor program and the installation program shall be provided for each AMG on two 5.25 inch, double density, MS-DOS formattable floppy diskettes. Each diskette shall be formatted to 360 Kilobytes using MS-DOS version 3.0 or later. One diskette shall be labelled "AMG PC software, diskette #1" and shall contain the following files: "install.bat", "install1.bat", and "amg.exe". The second diskette shall be labelled "AMG PC software, diskette #2" and shall contain the files "amg.cip", "amg.cnf", "amg.dtl", "autoexec.bat", and "config.sys". The two diskettes shall be exact copies of the Government Loaned Property diskettes (see 3.11) and shall be prepared and tested in accordance with appendix test method 2. Each set of prepared diskettes shall be able to install a working alarm-monitor program on an MS-DOS computer.

3.6 Burn-in. Each AMG shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Reliability. The AMG shall demonstrate a mean-time-between-failure (MTBF) of 2,000 hours or greater when tested as specified in 4.5.2.10.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Identification marking. The AMG shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10 Workmanship. Workmanship shall be in accordance with the requirements and guidance paragraphs of MIL-STD-454, requirement 9.

3.10.1 Soldering. All soldering shall be performed in accordance with MIL-STD-2000.

3.10.2 Welding. Sheet metal welding shall be in compliance with AWS D 9.1.

3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government

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(see 6.7) for manufacturing and testing the AMG and major components when furnished separately:

Item No.	Description	Identification	Quantity
1.	Alarm-monitor Program Diskette	AMG PC Software, 13229E8349, diskette #1	1
2.	Alarm-monitor Program Diskette	AMG PC Software, 13229E8349, diskette #2	1
3.	AMG functional test setup diskette	AMG functional test setup diskette	1
4.	Magnetic diskette Object Code, CMA	13229E8325	1
5.	Mylar drawing of PWB, Backplane, Right Hand	13229E8147 Sheet 3, 4, 5, 6	1
6.	Mylar drawing of PWB, Backplane, Left Hand	13229E8192 Sheet 3, 4, 5, 6	1
7.	Mylar drawing of PWB, I/O	13229E8196 Sheet 2, 3	1
8.	Mylar drawing of PWB, CPU	13229E8171 Sheet 2, 3, 4	1
9.	Mylar drawing of PWB, Surge Suppressor	13229E8336 Sheet 2, 3, 4	1
10.	Transmitter, Data, T-1257/FSS-9(V)	NSN 6350-00-251-5749	16
11.	Mylar drawing of PWB, No. 4	13220E3552 Sheet 2, 3	1
12.	Code Plug Set, 13220E3607-1	NSN 6350-01-071-5520	16
13.	Personal Computer, ISA bus, with EGA Monitor and Dot-Matrix Printer		3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspection. The AMG shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. Each first article AMG shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article AMG shall be tested as specified in 4.5.2.1 through 4.5.2.10, as applicable. Failure of any test shall be cause for rejection.

4.3.2.1 AMG. The AMG shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.3.2.2 Measurement recording requirements. All measurements, including voltages, currents, resistances, times, and frequencies observed during first article tests shall be recorded on the test data sheets. All measurements shall be to the precision necessary to demonstrate compliance with requirements in the appendix test method being conducted.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table II-A.

4.4.2 Examination.

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4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. A lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.4.3 Tests.

4.4.3.1 Individual. Each AMG shall be subjected to the tests in table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested in accordance with column 3 of table I. A lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The AMG shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the AMG and major components when furnished separately (see 3.3.2).
106. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified (see 3.1.3).
107. Components missing or not as specified (see 3.1.1).
108. Assembly not correct. (see 3.1).
109. Identification or special marking missing, incomplete or illegible (see 3.9).
110. Interchangeability not as specified (see 3.8).
111. Workmanship including wiring and soldering not as specified (see 3.10).
112. Welding not in accordance with AWS D 9.1 (see 3.10.2).
113. Soldering not in accordance with MIL-STD-2000 (see 3.10.1).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient humidity conditions at room ambient temperature of 76 ±8 °F. After successful completion of a functional test, each AMG shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the AMG shall operate as specified herein without maintenance.

4.5.2.2 Test schedule.

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TABLE I. Test schedule for the AMG.

First Article	Quality Conformance		Test	Test Para.	Reqt. Para.
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	4.5.2.5	3.5
X	X	-	AMG Program Diskette Prep & Test	4.5.2.6	3.5.4
X 1/	X 1/	-	CMA Input Sensitivity	4.5.2.7	3.5.1.4
X 1/	X 1/	-	Tamper Switch Tolerance	4.5.2.8	3.5.1.5
X	-	X	Emergency Power	4.5.2.9	3.5.3
X	-	-	Reliability	4.5.2.10	3.7

1/ This test need only be performed one time, during the first functional test after burn-in.

4.5.2.3 Test methods. Tests shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each AMG shall be subjected to a 24-hour burn-in after successful completion of a functional test (see 4.5.2.5). During the burn-in, the AMG shall be subjected to appendix test method 4. Appendix test method 1 shall be performed after completion of burn-in. Non-compliance with 3.5.1 shall constitute a failure. Any failure which is attributable to the commercial PC or its components shall not be charged as a burn-in failure against the AMG. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the functional test.

4.5.2.5 Functional. The AMG shall be tested in accordance with appendix test method 1. Nonconformance to 3.5.1 shall constitute failure of the test.

4.5.2.6 AMG program diskettes preparation and test. The AMG program diskettes shall be prepared and tested in accordance with appendix test method 2. Nonconformance to 3.5.4 shall constitute failure of the test.

4.5.2.7 CMA input sensitivity. The CMA shall be tested in accordance with appendix test method 6. Nonconformance to 3.5.1.4 shall constitute failure of this test.

4.5.2.8 Tamper switch tolerance. The CMA tamper switches shall be tested in accordance with appendix test method 3. Nonconformance to 3.5.1.5 shall constitute failure of this test.

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4.5.2.9 Emergency power. The UPS shall be tested in accordance with appendix test method 5. Nonconformance to 3.5.3 shall constitute failure of the test.

4.5.2.10 Reliability. Using an upper test MTBF θ_0 of 2,000 hours and a lower test MTBF θ_1 of 1,000 hours, the AMG shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-HDBK-781, test plan VIIID. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. Any failure which is attributable to the commercial PC or its components shall not be charged as an AMG failure. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.5.2.10.1 Test procedure. One or more AMGs shall be randomly selected by a Government representative. The testing shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Equipment shall be energized at the power input specified, with the specified batteries installed. Operation shall be continuous throughout the test, except as required during periodic performance checks. These checks shall be performed not less than nine times during each 24-hour period, but not more often than once every 45 minutes. This exercise shall consist of performing test method 4. The test shall commence only after successful completion of functional testing and a burn-in period of 24-hours, as specified in 4.5.2.4. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. No adjustments shall be allowed during the test.

4.5.2.10.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects listed in 4.6.2.3. Any nonconformance to the packaging requirements shall be cause for rejection of the first article. The deficiencies shall be corrected and the item reexamined for compliance. Level A pack of CMA, UPS, and batteries shall be subjected to free fall drop test as described in FED-STD-101.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

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4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

113. CMA not preserved and packed in a box as specified for level A (see 5.2.1.1 and 5.3.1.1).
114. UPS not preserved and packed in a box as specified for level A (see 5.2.1.2 and 5.3.1.2).
115. CCCA not wrapped with ESD material and placed in a box as specified for level A (see 5.2.1.3) or commercial (5.2.2).
116. Alarm-Monitor Program diskettes not placed in envelopes of ESD material and placed in a box as specified for level A (see 5.2.1.4) or commercial (5.2.2).
117. UPS cable assembly and RS-485 cable assembly not coiled and preserved in a box as specified for level A (see 5.2.1.5 and 5.2.1.6).
118. Keyboard templates not preserved as specified for level A (see 5.2.1.7).
119. Batteries not preserved and packed in a box as specified for level A (see 5.2.1.8 and 5.3.1.3).
120. Major components not consolidated in a box and packed as specified for level A (see 5.2.1.9 and 5.3.1.1).
121. Quantities packed together exceed the size or weight specified for level A (see 5.3.1.1).
122. Strapping, blocking, bracing, cushioning not as specified for level A (see 5.3.1.1).
123. Components procured separately, not packed as specified for level A (see 5.3.1.4.1 - 5.3.1.4.5).
124. Commercial preservation and packing not in accordance with the referenced document (see 5.2.2 and 5.3.2).
125. Marking missing, illegible, incorrect, or incomplete for levels A or commercial (see 5.4).

5. PACKAGING.

5.1 First article pack. Unless otherwise specified, the contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the requirements of this subject specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be accomplished utilizing either the first article or a production AMG. If the first article model AMG is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article AMG and the production AMG.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each AMG and major components when furnished separately shall be preserved as follows:

5.2.1.1 Converter Multiplexer Assembly (CMA). Each CMA shall be preserved in accordance with MIL-P-116, method IIb, except that the inner container shall be omitted. The container shall conform to PPP-B-636, type CF, class WWVR, variety

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DW, V11c. The CMA shall be cushioned and secured within the container to prevent movement or damage.

5.2.1.2 Uninterruptible Power Supply (UPS) without batteries. Each UPS shall be preserved as specified in 5.2.1.1.

5.2.1.3 Communications Circuit Card Assembly (CCCA). Each CCCA shall be completely wrapped or placed in a heat sealed bag, as required, of ESD material in accordance with MIL-P-116. If wrapped, the wrap shall be secured with tape in accordance with PPP-T-60, type III, class 1. The wrapped or bagged CCCA shall then be placed in a close-fitting box conforming to PPP-B-636, grade W6c, style optional. The CCCA shall be cushioned within the box to prevent movement or damage.

5.2.1.4 Alarm Monitor Program Diskettes. The two alarm monitor program diskettes (diskettes #1 and #2), each enclosed in their sleeves, shall be placed in an envelope lined with ESD material in accordance with MIL-P-81997. The enveloped diskettes shall then be placed in a close-fitting box conforming to PPP-B-636, grade 125, style optional.

5.2.1.5 Uninterruptible power supply cable assembly. Each UPS cable assembly shall be loosely coiled, without causing kinks or damage, and preserved in accordance with MIL-P-116, method IC-1 or IC-3. The cable assembly shall be cushioned at the ends and secured within the bag. The bagged cable assembly shall be placed in a close-fitting box conforming to PPP-B-636, grade 125, style optional.

5.2.1.6 RS-485 cable assembly. Each RS-485 cable assembly shall be preserved as specified in 5.2.1.5.

5.2.1.7 Keyboard Templates. The two keyboard templates shall be preserved together in a bag conforming to MIL-P-116, method IC-1 or IC-3. Bagged templates shall then be placed into a close fitting box conforming to PPP-B-636, grade 125, style optional.

5.2.1.8 Battery. Each battery shall be placed in a close-fitting box conforming to PPP-B-636, type CF, variety SW or DW, grade V3c. Cushioning shall be used to prevent movement or damage. Box closure shall be in accordance with the appendix to the box specification.

5.2.1.9 Consolidation. The UPS cable assembly, RS-485 cable assembly, keyboard templates, CCCA, and the diskettes, preserved as specified in 5.2.1, shall be consolidated in a close-fitting box conforming to PPP-B-636, type CF, W5c, style optional. The contents shall be cushioned and secured within the box to prevent movement or damage.

5.2.2 Commercial. Each AMG and the major components shall be preserved in accordance with ASTM D 3951 except the electronically sensitive items shall be preserved as specified for level A.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A.

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5.3.1.1 Converter Multiplexer Assembly (CMA) and consolidated components. The CMA, preserved as specified in 5.2, and the consolidated components as specified in 5.2.1.9, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the size and weight limitation of the box. Blocking, bracing, and cushioning shall be used to prevent movement and isolate impact shock as specified in MIL-STD-1186. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.3.1.2 Uninterruptible Power Supply (UPS). The UPS, preserved as specified in 5.2, shall be packed as specified in 5.3.1.1.

5.3.1.3 Battery. Six batteries, preserved as specified in 5.2, shall be packed together as specified in 5.3.1.1.

5.3.1.4 Components procured separately.

5.3.1.4.1 CMA and UPS. The CMA and UPS preserved as specified in 5.2, shall be packed individually as specified in 5.3.1.1 and 5.3.1.2 respectively.

5.3.1.4.2 Batteries. The batteries preserved as specified in 5.2, shall be packed together as specified in 5.3.1.3.

5.3.1.4.3 CCCA. One or more CCCAs preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-636, type CF, class WWVR, in quantities not to exceed the size and weight limitations of the box. Box closure shall be as specified by the appendix to the box specification.

5.3.1.4.4 Cable assemblies. One or more UPS cables or RS-485 cables preserved as specified in 5.2, shall be packed as specified in 5.3.1.4.3.

5.3.1.4.5 Keyboard templates. One or more sets of keyboard templates preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-636, grade W6s or W5s, style optional.

5.3.2 Commercial. Each AMG and major components or any components procured separately, preserved as specified in 5.2, shall be packed in accordance with ASTM D 3951.

5.4 Marking. Marking shall be level A or commercial as specified (see 6.2).

5.4.1 Level A. In addition to any special marking specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking shall be in accordance with ASTM D 3951. In addition, each shipping container shall display its gross weight and cube.

5.4.3 Special Marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AMG is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
c. Nomenclature and quantity of item(s) to be furnished.
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval, the number of units required (see 3.2 and 4.3).
f. When Government-furnished property is not required (see 3.11).
g. Degree of preservation and packing required (see 5.2 and 5.3).
h. When a first article pack is required (see 5.1).
i. Schedule, content, and format of burn-in failure reports (see 4.5.2.1).
j. Any special marking required (see 5.4.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First article pack. Any changes or deviations of a production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the AMG in accordance with this specification.

6.5 Definitions. The following definitions shall apply throughout this specification.

6.5.1 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the AMG; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

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6.6 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and test performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11 when applicable (see 6.2).

6.9 Subject term (key word) listings.

Intrusion detection
Physical security

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA-GS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6350-0133

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APPENDIX

FOR

ALARM-MONITOR GROUP

10. Scope. The test methods contained in this appendix are to determine whether the AMG and its major components procured under this purchase description conform to the requirements set forth. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

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TEST METHOD NO. 1

AMG FUNCTIONAL TEST

1. TEST APPARATUS:

(a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
(b) 16 J-SIIDS data transmitters (T-1257/FSS-9(V)) with code plugs (13220E3607-1) installed.
(c) Power supply - Hewlett-Packard 6202B, or equivalent.
(d) Diskette labelled "AMG functional test setup diskette".
(e) Alarm-status simulators.
(f) Ground strap.

2. TEST PROCEDURE:

2.1 Setup.

(a) Assemble the AMG components and the PC in accordance with figure 1. Place the appropriate keyboard template on the computer keyboard adjacent to the function keys.
(b) Connect the data transmitters to the test fixtures in accordance with figure 2. Each output shall be connected in parallel to four other pairs of wire which connect to separate inputs on the CMA (each data transmitter shall simulate 4 remote areas). The data transmitter outputs shall be connected to the CMA in accordance with table II.
(c) Connect the alarm-status simulators to the data transmitters in accordance with figure 2. Set switches SW1, SW2 and SW4 to the "closed" position and switch SW3 to the "open" position.
(d) Configure the CCCA in accordance with 3.5.2. Open the case of the PC as stated in the PC manufacturer's instructions. Install the CCCA into the PC in accordance with the manufacturer's instructions. Replace the cover on the PC.

CAUTION: Wear a grounding strap when handling CCCA.

(e) Connect the communications cable to the CCCA and the CMA. Connect the UPS status cable to the UPS and the CMA. Plug the CMA and PC components into the UPS AC power outlet.
(f) Power on the UPS and the PC. Using diskettes created in accordance with test method 2, section 2.1, install the AMG program onto the PC as described in test method 2, section 2.3. The AMG program shall start up automatically when the computer is restarted.

2.2 Verify communications between the CMA and the PC.

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(a) When the AMG program has started, press <F10>, type "setup" in lower case letters, and then press <ENTER>. Press "2" and <ENTER>. The status indication for one module shall be highlighted.
(b) Press the space bar once to change the status indication to "ON". Within 3 seconds, under the heading "A4 CCA's" the display shall show "YYYYYYYY", and under the heading "SOFTWARE VERSION" the display shall show "1.2".
(c) Press the down arrow key to move the highlighting to the next module status indication. Repeat step b until all 8 modules have been brought online this way.
(d) Press <ESC> and then "6" and <ENTER>. Press <F4> to view the Monitor Area Status screen. All 8 module status indications shall display "ON" and no alarms.
(e) Disconnect the communications cable between the CMA and the computer from the back of the computer. Blinking white-on-red "COM" alarms shall be displayed for all 8 modules and the printer shall print "COM-FAIL" messages for each module. Press <F1> to silence the beeper and stop the indicators from blinking.
(f) Re-connect the communication cable to the computer. All 8 module "COM" indicators shall change to gray-on-red.
(g) The printer shall print a "Com-norm" message for each of the 8 modules.
(h) Press <F2> to reset and clear the "COM" alarms. The printer shall print the reset of the "COM" for all 8 modules.

2.3 CPU card slots.

(a) Unplug the CMA power. Press <F1> to silence the beeper. Open the top drawer of the CMA. Clip a ground strap to an unpainted metal part of the chassis and wear the ground strap.
(b) As viewed from the drawer side of the CMA, remove the CPU card on the left side toward the front of the drawer, then remove the CPU card on the right side toward the back of the drawer. Close the top drawer and open the bottom drawer. Remove the two CPU cards from the same positions in that drawer. Close the bottom drawer and restore power to the CMA.
(c) All the even-numbered modules (2, 4, 6, 8) shall indicate a gray-on-red "COM" and the odd-numbered modules shall indicate a white-on-red "COM" .
(d) Press <F2>. All of the gray "COM" alarms shall disappear. White-on-red "COM" alarms shall remain.
(e) Turn off power to the CMA. Press <F1> to silence the beeper. Replace the removed CPU cards and remove the other CPU CCAs. Close the drawers and restore power.
(f) All the odd-numbered modules (1, 3, 5, 7) shall indicate a gray-on-red "COM" and the even-number modules shall indicate a white-on-red "COM".
(g) Turn off power to the CMA and reinstall the CPU cards. Press <F1> to silence the beeper. Close the drawers and power on the CMA.
(h) Press <F2>. All of the "COM" indications shall disappear.

2.4 Verify communications from the data transmitters to the PC.

(a) Press <F10>, type "setup" in lower case and press <ENTER>, type "7" and press <ENTER> again. Type "Y" and press <ENTER>.
(b) Insert the diskette labelled "AMG functional test setup diskette, version 1.0" into drive "a:". Type "copy a:amg.cnf c:\amg" then press <ENTER>.

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Press the key combination <ctrl-alt-delete>. The AMG program shall start automatically within one minute.

(c) Press <F5>. Each of the 32 (1 -32) displayed areas shall display "SEC AC" in gray and "SYNC" in light blue. Press <page down>. Each of the 32 (33-64) displayed areas shall display "SEC AC" in gray and "SYNC" in light blue. Press <page up>. Within 5 minutes all the areas shall display "SEC AC" in green and shall not display "SYNC". Press <page down> and <page up> as needed to view all the areas.

(d) Using the alarm-status simulator for a data transmitter, switch SW2, SW3, and SW4. Press <F7>.

NOTE: Each change shall cause a change on 4 areas. Reference Table II for the areas assigned to each data transmitter.

Four areas shall be displayed on the screen within 5 seconds. Each area's status shall indicate a flashing yellow "ACC", a flashing yellow "BAT" and flashing white-on-red "ALM". The beeper shall sound.

(e) Press <F1>. The flashing indication for one area shall stop and the beeper shall stop.

(f) If the green pointers do not point at the top area on the list, press <HOME>. Press the down arrow key and <ENTER>. Press <F1>. The flashing indication shall stop for the next area. Repeat this step until there are no flashing indications displayed on the screen. Press <ESC>.

(g) Switch the alarm-status simulator switches back to the starting position. Each of the same four areas shall have the following indication within 5 seconds: "SEC" displayed in flashing green; "AC" in flashing green; "ALM" in solid gray on red; the beeper shall sound. Press <F1> and the flashing for one of the areas shall stop and the beeper shall silence. Press <F2> and the alarm indication for that area shall disappear. Press <ESC> and the area shall be removed from the screen.

(h) Press <ENTER>. Press <F1> and then <F2>. Press <ESC> and the area shall be removed from the screen. Repeat this step for the last two areas.

(i) Repeat steps d, e, f, g, and h for the remaining 15 data transmitters. Alarms and status changes shall be reported on all 64 areas.

(j) Open switch SW1 on the alarm-status simulator for one data transmitter. The 4 corresponding areas shall display "COM" in flashing white-on-red. Press <F1> to silence the beeper.

(k) Press the switch SW5 on the alarm-status simulator for that data transmitter. Close switch SW1. Within 5 minutes the "COM" indications for each of the four areas shall change to gray-on-red and the printer shall indicate "Resynch-A" for each of the four areas. Press <F2> and then <ESC>. One area shall be removed from the list.

(l) Press <ENTER>, <F1>, then <F2>. This shall remove the next area from the screen. Repeat for the last two areas.

(m) Repeat steps j, k, and l for the remaining 15 data transmitters.

(n) Open switch SW1 on the alarm-status simulator for one data transmitter. The 4 corresponding areas shall display "COM" in flashing white-on-red. Press <F1> to silence the beeper.

(o) Press the switch SW5 on the alarm-status simulator for that data transmitter. Close switch SW1. Press the switch SW5 again for that data transmitter. Within 5 seconds, the "COM" indications for each of the 4 areas shall change to gray-on-red. The printer shall indicate "Resynch-M" for each of the 4 areas. Press <F2> and then <ESC>.

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(p) Press <ENTER>, <F1>, then <F2>. This shall remove the next area from the screen. Repeat for the last two areas. There shall be no areas displayed on the screen.
(q) Repeat steps n, o, and p for the remaining 15 data transmitters.

2.5 Data receiver card slots.

(a) Remove power from the CMA and open the top drawer. Press <F1> to silence the beeper. Remove every other data receiver CCA, starting with the CCA at the left front of the drawer and moving in a clockwise direction as viewed from above. Close the top drawer and open the bottom drawer. Remove every other data receiver CCA from the bottom drawer as done with the top drawer. Close the bottom drawer and plug in the CMA power.
(b) Press <F5>. Each of the displayed areas shall display a "COM" in blinking white-on-red and "SEC AC" in gray. Press <F1> to silence the beeper. Within 5 minutes the "COM" indication for each of the even-numbered areas (areas 2, 4, 6, etc.) shall change to gray-on-red and the "SEC AC" indications for those area shall be green. Press <page down>. The same shall be true for the even-numbered areas 34 through 64. Press <page up>. The printer shall print "Resynch-A" for all of the even-numbered areas. None of status indications for the odd-numbered areas shall change.
(c) Press <F10>, type "setup". Press "2" and <ENTER>. Under the heading "A4 CCA's", there shall be a "Y" for each module under the even-numbered columns, but no "Y" in any of the odd-numbered columns. Press <ESC>, then "6" and <ENTER>.
(d) Remove power from the CMA. Press <F1> to silence the beeper. Replace the data receiver CCAs and remove the other data receiver CCAs. Power the CMA back on. Press <F1> to silence the beeper.
(e) Within 5 minutes all the "COM" indications for each of the odd-numbered areas (areas 1, 3, 5, etc.) shall change to gray-on-red and "SEC AC" shall change to green. The printer shall print "Resynch-A" for all of the odd-numbered areas. Each of the even-numbered areas shall continue to indicate a white- on-red blinking "COM" and the status indicators for those areas shall be gray. Press <page down>. The same shall be true for the odd-numbered areas 33 through 63. Press <page up>.
(f) Press <F10>, type "setup". Press "2" and <ENTER>. Under the heading "A4 CCA's", there shall be a "Y" for each module under the odd-numbered columns, but no "Y" in any of the even-numbered columns. Press <ESC>, then "6" and <ENTER>.
(g) Turn off power to the CMA. Press <F1> to silence the beeper. Replace the data receiver CCAs. Power the CMA back on. Press <F1> to silence the beeper.

2.6 Computer powered on separately.

(a) Remove power from the computer, monitor, and printer. Wait approximately 10 seconds and restore power to the computer equipment.
(b) The AMG program shall restart automatically. When the initialization screen goes away after approximately 10 seconds, press <F4>. The status indicators for modules 1-8 shall display "ON" and there shall be no "COM" alarms displayed.

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2.7 CMA capacity.

(a) Press <F5>. Simultaneously switch SW2 on all 16 alarm-status simulators.
(b) Within 12 seconds the number under the heading "IN-ALARM" shall be "64" in blinking white-on-red. Each area displayed on the screen (1-32) shall display a blinking white-on-red "ALM" indication. Press <page down>. Each area displayed (33-64) shall display a blinking white-on-red "ALM" indication.

2.8 Tamper alarm.

(a) Assure that the CMA door, drawers, panel, and vent are closed securely. Press <F4>. The "Interface Rack" status shall indicate "SECURE" in green.
(b) Open the door on the CMA. The "Interface Rack" status shall indicate a blinking white-on-red "TAMPER" and the printer shall print a tamper alarm message. Close the door. The "Interface Rack" status shall change to a gray-on-red "TAMPER". Press <F1> and <F2>. The "Interface Rack" status shall change to a green "SECURE".

2.9 UPS status.

(a) The UPS status shall indicate "AC POWER" in green. Remove the input power to the UPS. The UPS status shall change to "BATTERY" in flashing yellow and the beeper shall sound. Press <F1>. The beeper shall silence and the "BATTERY" indication shall stop flashing.
(b) Restore power to the UPS. The UPS status shall change back to "AC POWER" in blinking green and the audible alarm shall beep. Press <F1>. The beeper shall silence and the "AC POWER" indication shall stop flashing.
(c) No other alarms shall be caused by switching the UPS.

2.10 Tamper and UPS reported by each module.

(a) Press <F10>, type "setup" and press <ENTER>. Press "2" and then <ENTER>. Press the down arrow key and then press the space bar. The status of module 2 shall change to "off". Repeat this until modules 2-8 all indicate "off". Press <ESC>, type "6" then press <ENTER>.
(b) Remove facility power from the UPS and open the back door of the CMA. Press <F4>. The UPS status shall indicate "BATTERY" in flashing yellow and the Interface Rack status shall indicate "TAMPER" in flashing white-on-red. Press <F1>. The beeper shall silence and the flashing indicators shall stop flashing.
(c) Restore facility power and close the CMA door. The UPS status shall change to "AC POWER" in flashing green and the Interface Rack status shall change to gray-on-red. The beeper shall beep. Press <F1> and then <F2>. "AC POWER" shall stop flashing, the Interface Rack status shall indicate "SECURE" in green, and the beeper shall stop beeping.
(d) Press <F10>, type "setup" and press <ENTER>. Press "2" and then <ENTER>. Press the down arrow key and then press the space bar. Using the up or down arrow keys, move the highlighting to the module whose status is "ON". Press the space bar. The status shall change to "Off". Press the down arrow key and press the space bar. The status of this module shall be "ON". Press <ESC>, type "6" and then press <ENTER>.

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(e) Repeat steps b, c, and d for the remaining modules. Each module shall report the UPS and Tamper status.

3. END OF TEST PROCEDURE:

NOTE: If all required test methods have been performed, proceed with this procedure. If not, perform this procedure after completing all required tests.

(a) Press <F10> and type "setup" in lower case. Press "7" then press <ENTER>; press "Y" then press <ENTER>.
(b) Insert the diskette titled "AMG functional test setup diskette" and type "a:uninstal" and <ENTER>. Follow any screen instructions.
(c) Turn off power to the computer equipment. Open the case of the computer and remove the CCCA from the card cage. Repack the circuit card and its manual in the original packaging.
(d) Disconnect cables from the CMA and the UPS.
(e) Disconnect AMG components from the test equipment; it is not necessary to disconnect test equipment from other test equipment (see figure 1).

TABLE II Data Transmitter Connections to CMA.

Data Transmitter	Area Nos.	CMA Connections			
1	1, 9, 17, 25	A1-B1,	A9-B9,	A1-B1,	A9-B9
2	2, 10, 18, 26	A2-B2,	A10-B10,	A2-B2,	A10-B10
3	3, 11, 19, 27	A3-B3,	A11-B11,	A3-B3,	A11-B11
4	4, 12, 20, 28	A4-B4,	A12-B12,	A4-B4,	A12-B12
5	5, 13, 21, 29	A5-B5,	A13-B13,	A5-B5,	A13-B13
6	6, 14, 22, 30	A6-B6,	A14-B14,	A6-B6,	A14-B14
7	7, 15, 23, 31	A7-B7,	A15-B15,	A7-B7,	A15-B15
8	8, 16, 24, 32	A8-B8,	A16-B16,	A8-B8,	A16-B16
9	33, 41, 49, 57	A1-B1,	A9-B9,	A1-B1,	A9-B9
10	34, 42, 50, 58	A2-B2,	A10-B10,	A2-B2,	A10-B10
11	35, 43, 51, 59	A3-B3,	A11-B11,	A3-B3,	A11-B11
12	36, 44, 52, 60	A4-B4,	A12-B12,	A4-B4,	A12-B12
13	37, 45, 53, 61	A5-B5,	A13-B13,	A5-B5,	A13-B13
14	38, 46, 54, 62	A6-B6,	A14-B14,	A6-B6,	A14-B14
15	39, 47, 55, 63	A7-B7,	A15-B15,	A7-B7,	A15-B15
16	40, 48, 56, 64	A8-B8,	A16-B16,	A8-B8,	A16-B16

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TEST METHOD NO. 2

AMG PROGRAM DISKETTES PREPARATION AND TEST

1. TEST APPARATUS:

   (a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
   (b) Master AMG program diskettes (two), labelled "AMG PC software, 13229E8349, diskette #1", and "AMG PC software, 13229E8349, diskette #2".

2. TEST PROCEDURE:

2.1 Create and verify diskettes.

   (a) Connect the AMG and computer equipment as in figure 1. Power on the equipment.
   (b) Use an unused, double-sided, double density, 5.25 inch diskette. For a computer with a single floppy disk drive, insert the blank diskette into the floppy disk drive. If the computer has two floppy drives, insert the blank diskette into drive "b:".
   (c) For a single drive computer, type the command: "format a: /4" and press <ENTER>. Follow screen directions to complete the command. For a two drive computer, type the command: "format b: /4" and press <ENTER>. Follow screen directions to complete the command.
   (d) When formatting is completed, remove the diskette and insert another blank diskette of the same type into the same drive. Press "y" then <ENTER>, and press <ENTER> again in response to the screen prompt to format another diskette. When the computer is finished formatting the second diskette, press "n" and <ENTER> in response to the prompt.
   (e) Insert the master diskette titled "AMG PC software, 13229E8349, diskette #1 into floppy diskette drive "a:".
   (f) For a single drive computer, type the command: "diskcopy a: a: /v" and press <ENTER>. Directions on the screen will prompt the user to switch diskettes. The master diskette will be called the "SOURCE" and the blank diskette will be called the "DESTINATION". Follow the screen directions. Each diskette will be inserted three times. For a two drive computer, type the command: "diskcopy a: b: /v" and press <ENTER>.
   (g) Create a diskette label with the title "AMG PC software, 13229E8349, diskette #1".
   (h) Remove the newly created diskette from the floppy disk drive and place the label on it. Place a write-protect tab over the small notch next to the label.
   (i) For a single drive computer, insert the master diskette titled "AMG PC software, 13229E8349, diskette #2" into the floppy disk drive. For a two drive computer, place the second blank diskette into drive "b:" and insert the master diskette titled "AMG PC software, 13229E8349, diskette #2" into drive "a:".
   (j) Follow the directions in step (f) to copy the diskette.

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(k) Create a diskette label with the title "AMG PC software, 13229E8349, diskette #2".
(l) When all of the files are successfully copied, remove the new diskette copy and place the identification label (see note 4, 13229E8349) and write-protect tab on it.

2.2 Verify diskettes.

(a) Insert the master diskette "AMG PC software, 13229E8349, diskette #1" into drive "a:". For a two drive computer, also insert the newly created diskette with the same title into drive "b:".
(b) For a single drive computer, type the command: "diskcomp a: a:" and press <ENTER>. Directions on the screen will prompt the user to switch diskettes. The master diskette will be called "SOURCE1" and the copy diskette will be called "SOURCE2". Follow the screen directions. Each diskette will be inserted twice. For a two drive computer, type the command: "diskcomp a: b:" and press <ENTER>.
(c) For a single drive computer, when the command is finished, remove the diskette and repeat steps (a) and (b) using the diskettes titled "AMG PC software, 13229E8349, diskette #2". For a two drive computer, the prompt "Do you want to compare more disks (Y/N)?" shall be displayed. Remove both diskettes and insert the master diskette labelled "AMG PC software, 13229E8349, diskette #2" into drive "a:", and insert the copy of it into drive "b:". Press "y" then <ENTER>. When finished, the same prompt shall appear. Press "n" then <ENTER> to complete the command. Remove the diskettes.

2.3 Install AMG monitor program from new copy.

(a) Place the newly copied diskette labelled "AMG PC software, 13229E8349, diskette #1" into drive "a:".
(b) Type "a:install" and press <ENTER>.
(c) When prompted by the message on the screen, insert the newly copied diskette #2 and press any key to complete installation.
(d) When installation is complete, remove the second diskette from the drive. Press the key combination <ctrl-alt-delete> to restart the computer.
(e) The AMG program shall start operating successfully.
(f) NOTE: If other tests are to be run at this time, do not proceed with this step.

Press <F10> , type "setup" then press <ENTER>. Type "7" and press <ENTER>, then press "y" and <ENTER>. This shall exit the AMG program.

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TEST METHOD NO. 3

TAMPER SWITCH TEST

1. TEST APPARATUS:

(a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
(b) Large Phillips screwdriver.
(c) Ruler.

2. TEST PROCEDURE:

(a) Setup the AMG and the test equipment in accordance with test method 1, section 2.1.
(b) Slowly open the door on the back of the CMA until the beeper on the PC sounds. Measure the distance the outer edge of the door travelled to cause the tamper alarm. See 3.5.1.5 for the allowable movement.
(c) "TAMPER" shall be indicated in white-on-red on the display and the beeper shall sound. Press <F1> to silence the beeper.
(d) Close the door. Press <F2>. The indicator shall change to a "SECURE".
(e) Loosen the screws on one of the drawers in the CMA. If necessary, slowly pull open the drawer. "TAMPER" shall be indicated on the display within the distance called out in paragraph 3.5.1.5.
(f) Close the drawer and tighten the screws. Press <F2>. The indicator shall change to "SECURE".
(g) Repeat step e and f for the second drawer.
(h) Loosen both screws on one side of the blank front panel on the CMA until a "TAMPER" is indicated. The panel movement shall cause the tamper alarm to be generated within the required distance.
(i) Tighten the screws. Press <F2>. The indicator shall change to "SECURE".
(j) Repeat steps h and i for the opposite end of the panel.
(k) Repeat steps h, i, and j for the ventilation panel on the CMA.

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TEST METHOD NO. 4

BURN-IN/RELIABILITY

1. TEST APPARATUS:

   (a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
   (b) 16 J-SIIDS data transmitters (T-1257/FSS-9(V)) with code plugs (13220E3607-1) installed.
   (c) Power supply - Hewlett-Packard 6202B, or equivalent.
   (d) Diskette labelled "AMG functional test setup diskette".
   (e) Alarm-status simulators.
   (f) Ground strap.

2. TEST PROCEDURE:

   (a) Setup the AMG in accordance with test method 1, 2.1.
   (b) Power on the equipment. The AMG program shall start running. Press <F10>, type "setup" in lower case, and press <ENTER>. Press "7" then press <ENTER>; press "y" then press <ENTER>.
   (c) Insert the diskette labelled "AMG functional test setup diskette" into drive a:. Type "copy a:amg.cnf c:\amg" and press <ENTER>. Press the key combination <ctrl-alt-delete>. The AMG program shall start automatically. When the program starts there shall be no alarms or status changes. Press <F5>.
   (d) Switch the alarm-status simulator switches SW2, SW3, and SW4 for each of the sixteen data transmitters. Each area on the display shall indicate "ACC", "BAT", and "ALM". This step shall be performed nine times during each 24 hour period at intervals of no less than 45 minutes.

NOTE: Each alarm and status change at the data transmitter shall cause a change on the monitor corresponding to the 4 areas. Reference table II for the areas assigned to each data transmitter.

   (e) Return the switches back to the original positions. The display shall indicate "SEC", "AC", and "ALM" for each area. Press <ESC>. Press <ENTER>, then press <F1> and <F2>. The indicators for area 1 shall not be blinking and the "ALM" indicator shall disappear.
   (f) Press the down arrow key, then press <ENTER>. Press <F1> and <F2>. Repeat this step for all 64 areas. When completed there shall be no flashing indications or "ALM" indications on the display.
   (g) The system shall be rejected if:

      (1) Erroneous communication failures are reported by the AMG,
      (2) Operator stimulated alarms and status changes are not reported within five seconds,
      (3) Erroneous alarms or status changes are reported by the AMG.

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TEST METHOD NO. 5

UPS CAPACITY

1. TEST APPARATUS:

   (a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
   (b) 16 J-SIIDS data transmitters (T-1257/FSS-9(V)) with code plugs (13220E3607-1) installed.
   (c) Power supply - Hewlett-Packard 6202B, or equivalent.
   (d) Diskette labelled "AMG functional test setup diskette".
   (e) Alarm-status simulators.

2. TEST PROCEDURE:
   (a) Setup test as in appendix test method 1, section 2.1.
   (b) Power up the AMG and PC for 12 hours or until the batteries are fully charged.
   (c) Disconnect facility power to the UPS and record the current time. Press <F1> to silence the beeper. The system shall continue to operate for at least 4 hours after removal of facility power.
   (d) After 4 hours, restore facility power to the UPS. The UPS batteries shall be fully charged in less than 24 hours.

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TEST METHOD NO. 6

CMA INPUT SENSITIVITY TEST

1. TEST APPARATUS:

   (a) Personal computer - IBM PC/AT compatible (ISA bus) with MS-DOS 3.0 or higher.
   (b) 16 J-SIIDS data transmitters (T-1257/FSS-9(V)) with code plugs (13220E3607-1) installed.
   (c) Power supply - Hewlett-Packard 6202B, or equivalent.
   (d) Diskette labelled "AMG functional test setup diskette".
   (e) 16 Alarm-status simulators.
   (f) Attenuator - Hewlett-Packard 350D, or equivalent.
   (g) Voltmeter, AC - Hewlett-Packard 400E, or equivalent.

2. TEST PROCEDURE:

   (a) Set up equipment as in figure 1 if not already done. The following steps shall be performed on a sample of 16 of the CMA inputs. A sample shall be taken from one input associated with each of the 16 data transmitters. See table II to reference input numbers to data transmitters.
   (b) If the program is running, press <F10>, type "setup" in lowercase, type "7" then press <ENTER>, type "y" and press <ENTER>. Whether the program was running or not, insert the diskette labelled "AMG functional test setup diskette, version 1.0" into drive "a:". Type "copy a:amg-j.cnf c:\amg-j" then press <ENTER>. Press the key combination <ctrl-alt-delete>. The AMG program shall start automatically within one minute. Within 5 minutes all of the areas shall be synchronized.
   (c) Remove data lines from the input to be tested. Connect these data lines to the attenuator pad. Connect a twisted pair of wires of no more than 10 feet long to the output of the attenuator and connect these lines to the CMA input to be tested. Set the attenuator to 0 dB attenuation.
   (d) Connect the ac voltmeter across the CMA input to be tested.
   (e) Wait for area associated with the input to resynchronize. Increase the attenuation in 1 dB steps, pausing 3 seconds between each step.
   (f) At sufficient attenuation, the display of the area associated with the input under test shall change to a "COM" alarm in flashing white-on-red, the audible alarm shall beep, and the printer shall a print "COM-FAIL" message.
   (g) Record the level displayed on the ac voltmeter. The level at which a "COM-FAIL" was reported shall be 35 ± 4 dBm.
   (h) Disconnect the attenuator lines from the input. Disconnect the data lines from the attenuator and connect them to the CMA input. The associated area shall resynchronize within 5 minutes.

   (i) Repeat steps (c) - (h) for each CMA input to be tested. If any of the inputs fail, all 64 CMA inputs shall be tested as described. Any A4 PWB failing shall be reworked as necessary for it to pass step (g).

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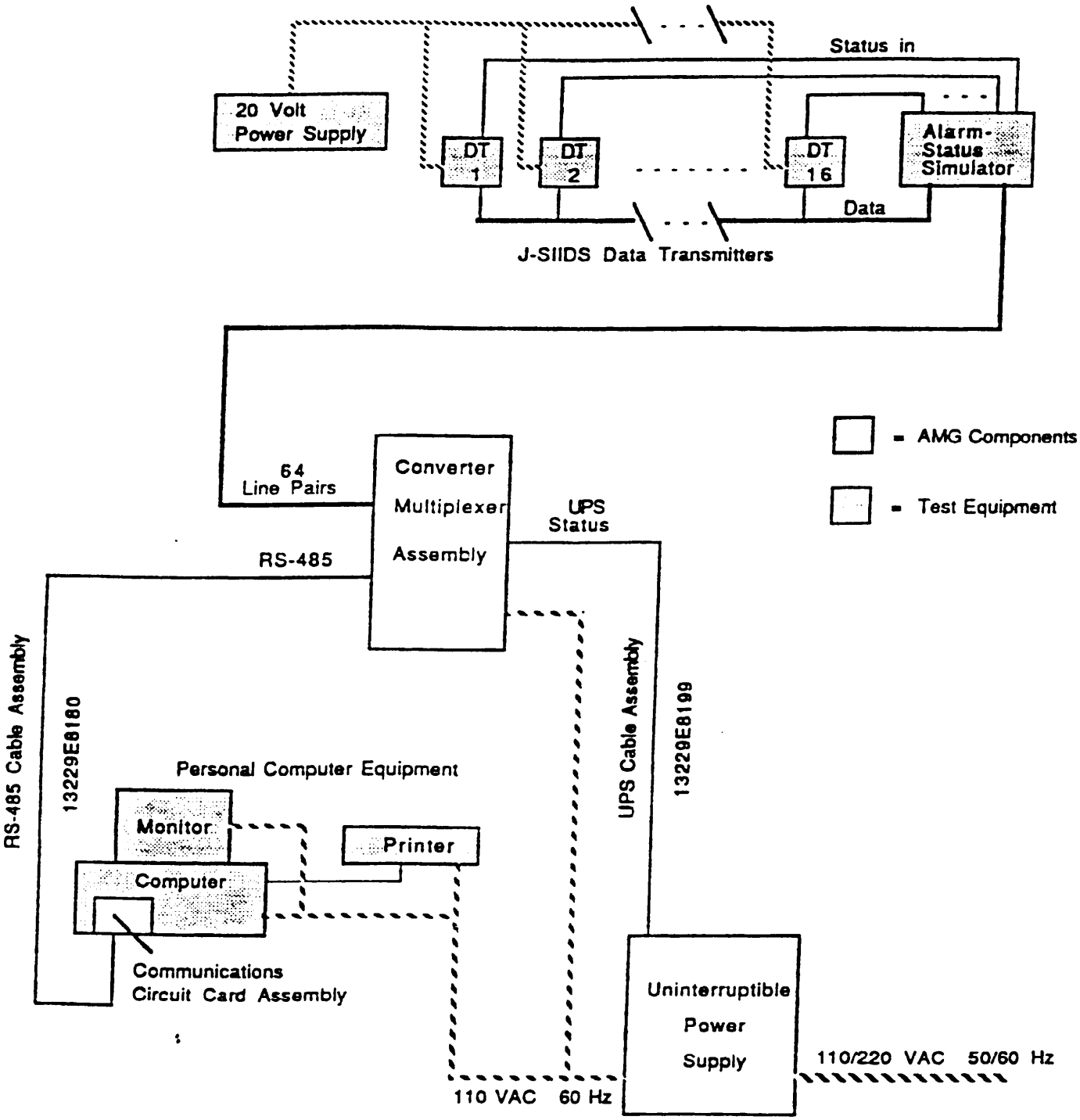


FIGURE 1. Alarm-Monitor Group acceptance test setup.

X-5030

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Typical 1 of 16

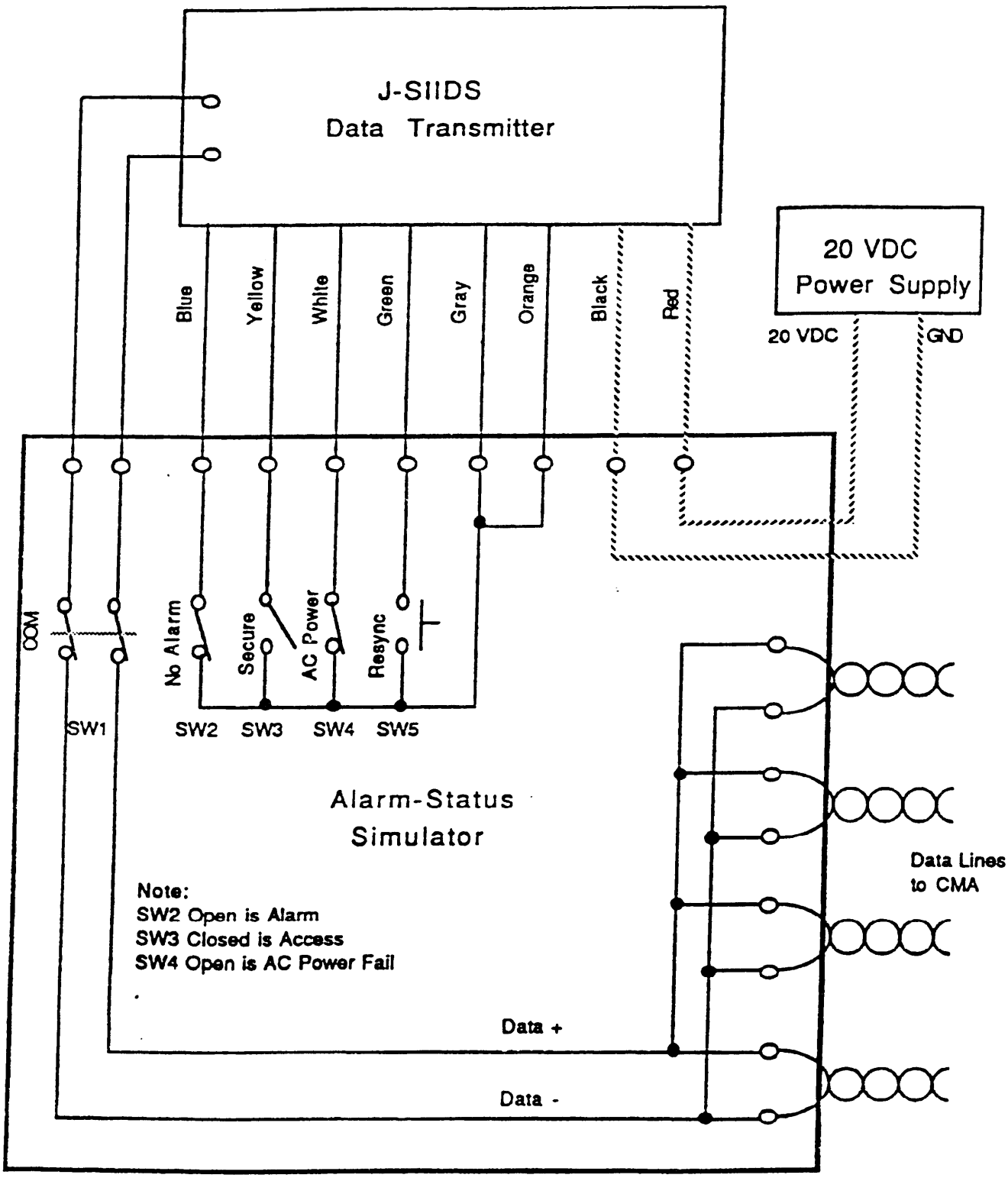


FIGURE 2. J-SIIDS data transmitter connections.

X-503.1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-53129	2. DOCUMENT DATE (YYMMDD) 920729

3. DOCUMENT TITLE
Alarm-Monitor Group OA-9431/FSS-9(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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